Lecture Two

1 Determinants and its Properties

1.1 Determinant

Definition 1.1.1 *The determinant of the matrix* $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ *is* $\det(A) = |A| = a_{11}a_{22} - a_{12}a_{21}$

Example 1.1.1 *1. For* $A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$, $\det(A) = |A| = \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(-3) = 4 + 3 = 7$

2. For $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$, $\det(B) = |B| = \begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 2(2) - 4(1) = 4 - 4 = 0$

3. For $C = \begin{bmatrix} 0 & \frac{3}{2} \\ 2 & 4 \end{bmatrix}$, $|C| = \begin{vmatrix} 0 & \frac{3}{2} \\ 2 & 4 \end{vmatrix} = 0(4) - 2\left(\frac{3}{2}\right) = -3$

1.2 Minors and Cofactors

To define the determinant of a square matrix of order higher than 2, it is convenient to use minors and cofactors.

Definition 1.2.2 *If* A *is a square matrix, then the minor* M_{ij} *of the entry* a_{ij} *is the determinant of the matrix obtained by deleting the ith row and jth column of A. The Cofactor* C_{ij} *of the entry* a_{ij} *is* $C_{ij} = (-1)^{i+j}M_{ij}$

For example, if A is 3×3 matrix, then the minors and cofactors of a_{21} and a_{22} are as shown below:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = a_{12}a_{33} - a_{13}a_{32}, \quad C_{21} = (-1)^{2+1}M_{21} = -M_{21}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} = a_{11}a_{33} - a_{13}a_{31}, \quad C_{22} = (-1)^{2+2}M_{22} = +M_{22}$$

The minors and cofactors of a matrix can differ only in sign. To obtain the cofactors of a matrix, first find the minors and then apply the check board pattern of +'s and -'s shown below:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Example 1.2.2 *Find all minors and cofactors of* $A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$

Sol

$M_{11} = \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = (-1)(1) - (0)(2) = -1$

$M_{12} = \begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = (3)(1) - (4)(2) = -4$

$M_{13} = \begin{vmatrix} 3 & -1 \\ 4 & 0 \end{vmatrix} = (3)(0) - (-1)(4) = 4$

$$M_{21} = \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} = (2)(1) - (0)(1) = 2$$
$$M_{22} = \begin{vmatrix} 0 & 1 \\ 4 & 1 \end{vmatrix} = (0)(1) - (4)(1) = -4$$
$$M_{23} = \begin{vmatrix} 0 & 2 \\ 4 & 0 \end{vmatrix} = (0)(0) - (4)(2) = -8$$
$$M_{31} = \begin{vmatrix} 2 & 1 \\ -1 & 2 \end{vmatrix} = (2)(2) - (-1)(1) = 5$$
$$M_{32} = \begin{vmatrix} 0 & 1 \\ 3 & 2 \end{vmatrix} = (0)(2) - (3)(1) = -3$$
$$M_{33} = \begin{vmatrix} 0 & 2 \\ 3 & -1 \end{vmatrix} = (0)(-1) - (3)(2) = -6$$

Now: we can write M as:

$$M = \begin{bmatrix} -1 & -5 & 4 \\ 2 & -4 & -8 \\ 5 & -3 & -6 \end{bmatrix}$$

Then by using $C_{ij} = (-1)^{i+j} \cdot M_{ij}$, we can write C as:

$$C = \begin{bmatrix} -1 & 5 & 4 \\ -2 & -4 & 8 \\ 5 & 3 & -6 \end{bmatrix}$$

1.3 Determinant of a Square Matrix

If A is a square matrix of order $n \geq 2$, then the $\det(A)$ is the sum of the entries in the first row of A multiplied by their respective cofactors. That is

$$\det(A) = |A| = \sum_{j=1}^{n} a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

Example 1.3.3 *Find the determinant of* $A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$

Sol

This is the same matrix as in example 1.2.2. There we found the cofactors of the entries in the first row to be $C_{11} = -1, C_{12} = 5, C_{13} = 4$. So, by the definition of a determinant we have:

$$|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = 0(-1) + 2(5) + 1(4) = 14.$$

Although the determinant is defined as an expansion by the cofactor in first row, it can be shown that the determinant can be evaluated by expanding in any row or column. For instance, we could expand the previous matrix in example 1.3.3 by the second row to obtain:

$$|A| = a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} = (3)(-2) + (-1)(-4) + (2)(8) = 14$$

Or by the first column to obtain

$$|A| = a_{11}C_{11} + a_{21}C_{2} + a_{31}C_{31} = (0)(-1) + (3)(-2) + (4)(5) = 14$$

Theorem 1.3.1 *Let A be a square matrix of order n, then the determinant of A is:*

$$\det(A) = |A| = \sum_{j=1}^{n} a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + a_{i3}C_{i3} + \cdots + a_{in}C_{in}$$

Or

$$\det(A) = |A| = \sum_{i=1}^{n} a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + a_{3j}C_{3j} + \cdots + a_{nj}C_{nj}$$

Note:

1. When expanding by cofactors, you do not need to find cofactor of zero entries because zero times its cofactor is zero.

2. The row (or column) containing the most zeros is usually the best choice for expansion by cofactors. The next example demonstrates this.

Example 1.3.4 *Find the determinant of* $A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}$

Sol

Notice that three of the entries in the third column are zeros. So, to eliminate some of the work in the expansion, use the third column.

$$|A| = 3C_{13} + 0C_{23} + 0C_{33} + 0C_{43}$$

The cofactors C_{23}, C_{33} *and* C_{43} *have zero coefficients, so we need only find the cofactor* C_{13}*. To do this delete the first row and third column of* A *and evaluate the determinant of the resulting matrix.*

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix}$$

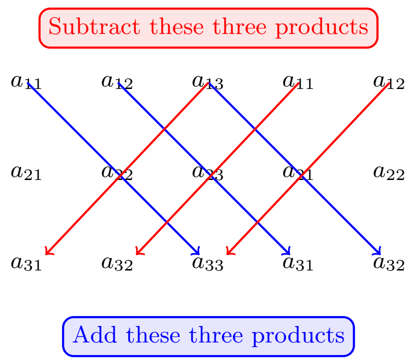
Now, expanding by cofactors in the second row yields

$$C_{13} = 0(-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 4 & -2 \end{vmatrix} + (2)(-1)^{2+2} \begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix} + (3)(-1)^{2+3} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} = 13$$

Now we obtain $|A| = 3(13) = 39$.

1.4 Sarrus' Rule

An alternative method is commonly used to evaluate the determinant of a 3×3 matrix A and called Sarrus' Rule. To apply this method copy the first and the second columns of A to form fourth and fifth columns. Then obtain the determinant of A by adding (or subtracting) the products of the six diagonals as shown in the diagram below:

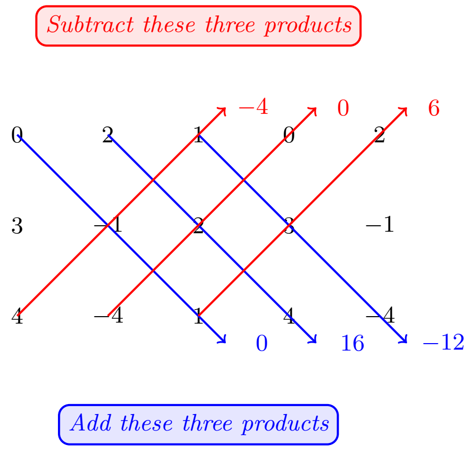


$$\Rightarrow |A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31}$$

Example 1.4.5 *Find the determinant of* $A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix}$

Sol

Begging by copying the first two columns and then computing the six diagonal products as shown velow:



By adding the lower three products and subtracting the upper three products, we can find the determinant of A to be:

$$|A| = 0 + 16 + (-12) - (-4) - 0 - 6 = 2$$

Note: The diagonal process illustrated in this example is valid only for matrices of order 3.

1.5 Determinant for Triangular Matrices

Example 1.5.6 *Find the determinant of* $A = \begin{bmatrix} 2 & 3 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$

Sol

By using the third row to expand, we obtain:

$$|A| = 0(-1)^{3+1}\begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix} + 0(-1)^{3+2}\begin{vmatrix} 2 & -1 \\ 0 & 2 \end{vmatrix} + 3(-1)^{3+3}\begin{vmatrix} 2 & 3 \\ 0 & -1 \end{vmatrix} = 3(1)(-2) = -6$$

Which is the product of the entries on the main diagonal.

Theorem 1.5.2 *If A is a triangular matrix of order n, then its determinant is the product of the entries on the main diagonal. That is:*

$$\boxed{|A| = a_{11} \times a_{22} \times a_{33} \times \cdots \times a_{nn}}$$

Example 1.5.7 *Find the determinant of* $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{bmatrix}$

Sol

$$|A| = (2)(-2)(1)(3) = -12$$

Exercise 1.5.1 *Solve for x.* $\begin{vmatrix} x+3 & 2 \\ 1 & x+2 \end{vmatrix} = 0$

1.6 Elementary Row Operations

1. **Replacement:** Replace one row by the sum of itself and a multiple of another row.
2. **Interchange** two rows.
3. **Scaling:** Multiplying all entries in a row by a non zero constant.

Theorem 1.6.3 *Let A be a square matrix, then:*

(a) If a multiple of one row of A is added to another row to produce a matrix B, then $\det(B) = \det(A)$

(b) If two rows of A are interchanged to produce B, then $\det(B) = -\det(A)$.

(c) If one row is multiplied by k to produce B, then $\det(B) = k.\det(A)$

Example 1.6.8 *Compute* $\det(A)$ *where* $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & 9 \\ -1 & 7 & 0 \end{bmatrix}$

Sol

$$\det(A) = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & 9 \\ -1 & 7 & 0 \end{vmatrix} \xrightarrow{R_2 \to R_2 + 2R_1} \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} \xrightarrow{R_3 \to R_3 + R_2} \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

$$= \xrightarrow[R_3 \to R_2]{R_2 \to R_3} \begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix}$$

An interchange of rows 2 and 3 reverses the sign of the determinant, so

$$\det(A) = -\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

Example 1.6.9 *Compute the determinant of A where* $A = \begin{bmatrix} 2 & -8 & 6 & 8 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix}$

Sol

$$A = 2\begin{bmatrix} 1 & -4 & 3 & 4 \\ 3 & -9 & 5 & 10 \\ -3 & 0 & 1 & -2 \\ 1 & -4 & 0 & 6 \end{bmatrix} \xrightarrow{\substack{R_2 \to R_2 + (-3)R_1 \\ R_3 \to R_3 + (3)R_1 \\ R_4 \to R_4 + (-1)R_1}} 2\begin{bmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & -12 & 10 & 10 \\ 0 & 0 & -3 & 2 \end{bmatrix} \xrightarrow{R_3 \to R_3 + (4)R_2} 2\begin{bmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$$= \xrightarrow{R_4 \to R_4 + \left(-\frac{1}{2}\right)R_3} 2\begin{bmatrix} 1 & -4 & 3 & 4 \\ 0 & 3 & -4 & -2 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now, $\det(A) = 2(1)(3)(-6)(1) = -36$

Exercise 1.6.2 *Given* $A = \begin{bmatrix} 0 & -7 & 14 \\ 1 & 2 & -2 \\ 0 & 3 & -8 \end{bmatrix}$, *find* $\det(A)$

Theorem 1.6.4 *If A is a square matrix and any one of the conditions below is true, then* $\det(A) = 0$.

1. *An entire row (or column) consists of zeros.*
2. *Two rows (or columns) are equal.*
3. *One row (or column) is a multiple of another row (or column).*

Example 1.6.10 *Find the determinant of* $A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{bmatrix}$

Sol

$$\det(A) = |A| = \begin{vmatrix} 1 & 4 & 1 \\ 2 & -1 & 0 \\ 0 & 18 & 4 \end{vmatrix} \xrightarrow{R_2 \to R_2 + (-2)R_1} \begin{vmatrix} 1 & 4 & 1 \\ 0 & -9 & -2 \\ 0 & 18 & 4 \end{vmatrix}$$

Now the second and third rows are multiplies of each other, so the determinant is zero.

1.7 Properties of Determinants

Example 1.7.11 *Find* $|A|$, $|B|$ *and* $|AB|$ *for the matrices* $A = \begin{bmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{bmatrix}$

Sol

$$|A| = \begin{vmatrix} 1 & -2 & 2 \\ 0 & 3 & 2 \\ 1 & 0 & 1 \end{vmatrix} = -7 \ , \ |B| = \begin{vmatrix} 2 & 0 & 1 \\ 0 & -1 & -2 \\ 3 & 1 & -2 \end{vmatrix} = 11 \ \text{ and } \ |AB| = \begin{vmatrix} 8 & 4 & 1 \\ 6 & -1 & -10 \\ 5 & 1 & -1 \end{vmatrix} = -77$$

Clearly, $|AB| = |A|.|B|$ *and it is true in general.*

Theorem 1.7.5 ***Determinant of a Matrices Product:*** *If* A *and* B *are square matrices of order* n*, then* $\det(AB) = \det(A).\det(B)$

Theorem 1.7.6 ***Determinant of a Scalar Multiple of a Matrix:*** *If* A *is a square matrix of order* n *and* c *is a scalar, then the determinant of* cA *is* $\det(cA) = c^n \det(A)$

Example 1.7.12 *Find* $\det(A)$ *if* $A = \begin{bmatrix} 10 & -20 & 40 \\ 30 & 0 & 50 \\ -20 & -30 & 10 \end{bmatrix}$

Sol

Given $A = 10 \begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{bmatrix}$ *and* $\begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = 5$. *So by applying theorem 1.7.6, we get* $|A| = 10^3(5) = 5000$.

Note: $|A| + |B| \neq |A + B|$.

Theorem 1.7.7 ***Determinant of a Transpose:*** *If* A *is a square matrix, then* $\det(A) = \det(A^T)$.

Exercise 1.7.3 *If* $A = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 0 & 0 \\ -4 & -1 & 5 \end{bmatrix}$. *Show that* $\det(A) = \det(A^T)$.

2 Inverse of a Matrix

2.1 Basic Concept

Definition 2.1.3 *An* $n \times n$ *matrix* A *is said to be* ***invertible*** *if there is an* $n \times n$ *matrix* C *such that*

$$CA = I = AC$$

Where $I = I_n$ *the* $n \times n$ *identity matrix. In this case,* C *is an* ***inverse*** *of* A.

In fact, C is uniquely determined by A, because if B were another inverse of A, then

$$B = BI = B(AC) = (BA)C = IC = C$$

This unique inverse is denoted by A^{-1}, so that

$$A^{-1}A = I \ \text{ and } \ AA^{-1} = I$$

A matrix that is not invertible is sometimes called (**singular matrix**), and and invertible matrix called non singular matrix.

Example 2.1.13 *If* $A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$ *and* $C = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$. *Then:*

$$AC = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$CA = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus $C = A^{-1}$

Theorem 2.1.8 *Let* $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, *if* $ad - bc \neq 0$, *then* A *is invertible and*

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Note: If $ad - bc = 0$, then A is singular.

Example 2.1.14 *Find the inverse of the matrix* $A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$.

Sol

To find the inverse of A*, solve the matrix equation* $AX = I$ *for* X*.*

$$\Rightarrow \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

By doing the matrices multiplication, we get the following equations

$$a + 4c = 1 \quad , \quad b + 4d = 0 \quad , \quad -a - 3c = 0 \quad , \quad -b - 3d = 1$$

and by solving these equations we get $a = -3, b = -4, c = 1, d = 1$. *i.e.*

$$A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$$

And you can check it by using matrix multiplication.

Generalizing the method used to solve example 2.1.14 provide a convenient method for finding an inverse. We will adjoin the given matrix with the identity matrix to obtain:

$$\begin{bmatrix} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{bmatrix}$$

Then we use row operations to put this matrix in the form $[I \quad B]$ where $B = A^{-1}$.

$$\begin{bmatrix} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_1 + R_2} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 \to R_1 + (-4) R_2} \begin{bmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 1 & 1 \end{bmatrix} \equiv [I \quad B]$$

Therefore $A^{-1} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$. **Note that:**

1. This procedure (or algorithm) works for an arbitrary $n \times n$ matrix.
2. If A cannot be row reduced to I_n, then A is not invertible (or singular). This procedure is called Gauss-Jordan Elimination.

Finding The Inverse of a Matrix by Gauss-Jordan Elimination:

1. Write the $n \times 2n$ matrix that consists of A on the left and $n \times n$ identity matrix I on the right to obtain $[A \quad I]$. This process is called adjoining matrix I to matrix A.
2. If possible, row reduce A to I using elementary row operations on the entire matrix $[A \quad I]$. The result will be the matrix $[I \quad A^{-1}]$. If this is not possible, then A is singular.
3. Check your work by multiplying to see that $AA^{-1} = I = A^{-1}A$

Example 2.1.15 *Find the inverse of the matrix* $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$

Sol

By adjoining the identity matrix to A to form the matrix $[A \quad I]$.

$$\begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + 6R_1]{R_2 \to R_2 + (-1)R_1} \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & 3 & 6 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \to R_3 + (4)R_2} \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 4 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 \to (-1)R_3} \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{bmatrix} \xrightarrow{R_2 \to R_2 + R_3} \begin{bmatrix} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 + R_2} \begin{bmatrix} 1 & 0 & 0 & -2 & -3 & -1 \\ 0 & 1 & 0 & -3 & -3 & -1 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{bmatrix}$$

Thus the matrix A is invertible and its inverse is:

$$A^{-1} = \begin{bmatrix} -2 & -3 & -1 \\ -3 & -3 & -1 \\ -2 & -4 & -1 \end{bmatrix}$$

Example 2.1.16 *Show that the following matrix has no inverse.*

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ -2 & 3 & -2 \end{bmatrix}$$

Sol

Adjoin the identity matrix to A to form $[A \quad I]$.

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ -2 & 3 & -2 & 0 & 0 & 1 \end{bmatrix}$$

Then apply Gauss-Jordan elimination to get: $\begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & -7 & 2 & -3 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{bmatrix}$. *So it is not possible to rewrite the matrix $[A \quad I]$ in the form $[I \quad A^{-1}]$. This means that A has no inverse (not invertible) or singular.*

2.2 Properties of Inverse

Theorem 2.2.9 *If A is an invertible matrix, k is a positive integer, and c is a non zero scalar, then A^{-1}, A^k, cA and A^T are invertible and the statements below are true.*

1. $(A^{-1})^{-1} = A$
2. $(A^k)^{-1} = A^{-1}.A^{-1}.A^{-1}.\dots.A^{-1} = (A^{-1})^k$
3. $(cA)^{-1} = \dfrac{1}{c}A^{-1}$
4. $(A^T)^{-1} = (A^{-1})^T$

Example 2.2.17 *Compute A^{-2}, two different ways and show that the results are equal.* $A = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$

Sol

$$A^2 = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 10 & 18 \end{bmatrix}$$

$$(A^2)^{-1} = \frac{1}{4} \begin{bmatrix} 18 & -5 \\ -10 & 3 \end{bmatrix} = \begin{bmatrix} \frac{9}{2} & -\frac{5}{4} \\ -\frac{5}{2} & \frac{3}{4} \end{bmatrix}$$

Another way, firstly we find A^{-1}, then we multiply it by itself to obtain $(A^{-1})^2$.

$$A^{-1} = \frac{1}{2}\begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$$

$$(A^{-1})^2 = \begin{bmatrix} 2 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 2 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{9}{2} & -\frac{5}{4} \\ -\frac{5}{2} & \frac{3}{4} \end{bmatrix}$$

Therefore, both methods produce the same result.

Theorem 2.2.10 *If A and B are invertible matrices of order n then, AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.*

Exercise 2.2.4 *Find $(AB)^{-1}$ for $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 3 \end{bmatrix}$ using $A^{-1} = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 0 \\ \frac{2}{3} & 0 & -\frac{1}{3} \end{bmatrix}$*

Theorem 2.2.11 ***Cancellation Properties:*** *If C is an invertible matrix then:*

1. *If $AC = BC \Rightarrow A = B$. (Right Cancellation)*
2. *If $CA = CB \Rightarrow A = B$. (Left Cancellation)*

Theorem 2.2.12 ***Determinant of an Inverse Matrix:*** *If A is an $n \times n$ invertible matrix, then*

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

Example 2.2.18 *Find $|A^{-1}|$ for $\begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$*

Sol

One way to solve this problem is to find A^{-1} then evaluate its determinant. But it is easier however to apply theorem 2.2.12 as shown below.

$$|A| = \begin{vmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{vmatrix} = 4$$

Now by using $\det(A^{-1}) = \dfrac{1}{\det(A)}$, we get $|A^{-1}| = \dfrac{1}{4}$.